

Preliminary Study of an Elastic Knee Exoskeleton Driver with Linearly Controllable Stiffness

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Abstract—Loss of knee articulation severely restricts elderly independence, making everyday sit-to-stand movements hazardous. Traditional rigid exoskeletons lack natural biomechanical compliance, failing to absorb impact forces effectively. This paper proposes a lightweight knee exoskeleton actuator featuring biomimetic, digitally linear variable stiffness. By employing three polyurethane (PU) elastic cords with binary-weighted cross-sectional areas (1:2:4) and an 8-angle cam-brake switching mechanism, the driver achieves discretized yet linearly controllable dynamic stiffness. Kinematic analysis demonstrates that incorporating an offset angle into the slide-link structure eliminates dual-solution instabilities and ensures smooth transmission throughout the knee's flexion range. Displacement and torque relationships were derived and experimentally evaluated via a dual-link leg simulator. The proposed binary-coded stiffness architecture provides an efficient, intrinsically safe approach to real-time exoskeleton stiffness tuning.

Keywords—knee exoskeleton; elastic actuator; variable stiffness

I. Introduction

Age-related sarcopenia and joint degeneration severely restrict knee articulation in older adults, making daily sit-to-stand (STS) transitions hazardous and significantly elevating fall risks [1]. Wearable knee exoskeletons capable of delivering targeted torque assistance present a vital clinical intervention to reduce muscular exertion and restore lower-limb mobility. Conventional exoskeleton designs predominantly rely on rigid drive mechanisms. Although rigid transmissions deliver high output torque, they lack intrinsic physical compliance, rendering the device incapable of absorbing high-frequency mechanical impacts and compromising natural human-robot interaction safety. While software-driven active impedance algorithms can emulate joint softness, their bandwidth limits prevent them from matching the instantaneous impact mitigation provided by physical elasticity. In biological systems, the musculoskeletal complex dynamically modulates joint torque and structural stiffness independently through the hierarchical recruitment of muscle motor units. Replicating this decoupled control architecture in artificial actuators is critical for adapting to multi-phase gait dynamics.

To embody physical compliance, various Variable Stiffness Actuators (VSAs) have been introduced for lower-limb exoskeletons. Schrade et al. [2] recognized the importance of variable stiffness in joint control; however, their architecture relied on manual pre-tensioning of springs across test conditions, limiting their system to offline pre-set validations rather than real-time dynamic stiffness control. Conversely, Yang et al. [3] modulated overall joint stiffness using a differential cam-spring mechanism. However, because the moment arm of the stretched springs varies continuously during joint rotation, the effective stiffness exhibits strong angular dependency—yielding high restorative forces at specific angles but minimal resistance at others. Consequently, existing VSA architectures struggle to achieve rapid dynamic switching, structural simplicity, and predictable linear stiffness mapping simultaneously. To address these limitations, this paper presents a lightweight knee exoskeleton actuator featuring biomimetic, digitally linear variable stiffness. By coupling a binary-weighted polyurethane (PU) elastic matrix with an 8-angle cam-brake selector, the proposed driver achieves discretized yet strictly linear dynamic stiffness modulation.

II. Overall System Design

The architecture presented in this work is inspired by biological observations of the human musculoskeletal -joint complex. Accordingly, the proposed knee actuator prioritizes the following functional characteristics:

1. Biomimetic Elasticity: Low-cost, commercial polyurethane (PU) cords are employed as compliance elements to mimic biological muscle tissue, providing passive compliance and intrinsic mechanical safety.
2. Hierarchical Recruitment: Actuation bundles are recruited sequentially in discrete stages, directly mirroring biological motor unit recruitment patterns.
3. Discretized Stiffness Modulation: System stiffness is governed dynamically via selective binary switching (in/out) of active PU elastic cords.
4. Linearization via Binary Coding: A binary-weighted arrangement is leveraged to achieve digitally linear dynamic stiffness regulation.

Specific Implementation

The proposed knee exoskeleton comprises three primary sub-assemblies:

1. A lower leg brace (LLB)
2. A 3-DOF hard-link kinematic structure (3DHL)
3. An elastic driver system, comprising:
 - o A Brushless DC (BLDC) motor
 - o A binary-coded PU elastic cord matrix
 - o A digitally controlled cam-brake system.

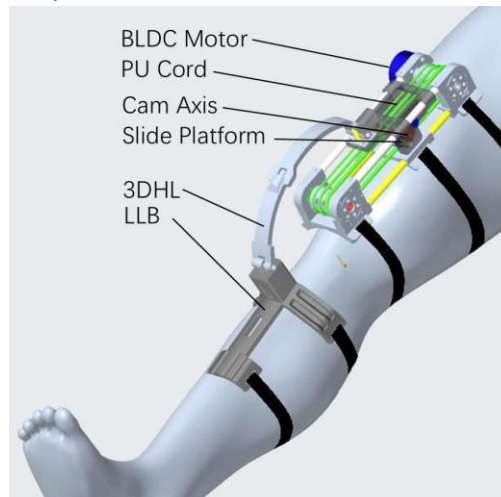


Fig. 1 - Overall Design of the Knee Exoskeleton

III. Detailed Design

A. Binary PU Cord System

In this research prototype, the PU elastic cord system includes 3 PU cords(85A), with diameters of 3mm, 4.24mm, and 6mm.

Table I - Binary Coding with PU Cord Diameter

PU Cord	Diameter (mm)	Area(mm ²)
Cord 1	3	9
Cord 2	4.24	18
Cord 3	6	36

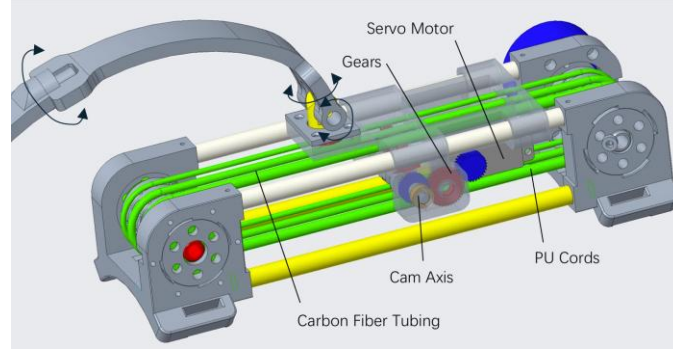


Fig. 2 - Detailed View of Elastic Driver

So the cross-section areas of these PU cords are 9, 18, and 36 mm², respectively. The ratio of these cross-section areas, and therefore of stiffnesses, is then 1:2:4. Because the stiffness is proportion to the area of the cross-section, the ratio of the stiffness is also 1:2:4.

B. Binary CAM Control

In order to control the switch in and out of these PU cords, a binary CAM is designed to implement binary coding with rotation angles. The CAM axis operates with 8 discrete angles: 0, 45, 90, 135, 180, 225, 270, 315 degrees. On the CAM axis there are 3 cams with high and low height positions. For each angle, the heights of these CAMs form a binary coding as shown below:

Table II - Binary Coding with Cam position

Angular position	Cam1 height	Cam2 height	Cam 3 height	Stiffness Coefficient
0	low	low	low	0
45	low	low	high	1
90	low	high	low	2
135	low	high	high	3
180	high	low	low	4
225	high	low	high	5
270	high	high	low	6
315	high	high	high	7

When a cam is high, it presses down on a brake on to a PU cord, binding the sliding platform to the PU cord.

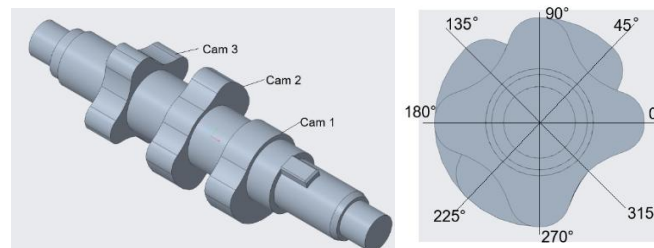


Fig. 3 - 8-Angle Cam System

As shown in Fig. 3, the cam axis is driven by a servo motor through gears. At a given angular position, the cams press down brakes on to their corresponding PU cords, engaging the PU cords with the sliding platform. When the height of a can is high, it presses down on the brake and connect the PU bank with the sliding platform; when the height is low, the brake is popped up by springs, which releases the PU cord from the sliding platform. The servo motor thus controls the switching in and out of these three PU cords by positioning the cam axis to one of

the 8 angles in Table II.

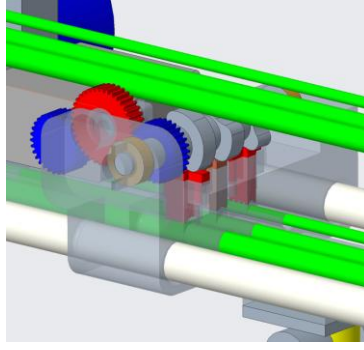


Fig. 4 Cam Controlled Brakes

In the mechanism shown in Fig. 2, the stiffness of the elastic driver without load is $K = \alpha EA/L$, where α is the stiffness coefficient as shown in Table II. E is Young's modulus of PU material. A is the cross-section area of the 3mm PU cord. L is the length of the PU cords from the motor wheel to the slide platform. When the cords are loaded, deformation has to be taken into consideration. A more accurate estimation should be

$$K = \alpha \cdot \frac{E_{tan}(F_0) \cdot A(F_0)}{L(F_0)} \quad (1)$$

E_{tan} is tangential modulus.

The complete mechanical design is made open-source and available on GitHub [6].

C. Mechanism Analysis

The knee exoskeleton is fundamentally an offset slider-crank mechanism as shown in Fig. 5. With the variables defined in Fig. 5, the lateral movement of the slide as a function of the rotation angle θ can be derived. The position of the end of L_1 is $(L_1 \sin \theta, -L_1 \cos \theta)$. Define the rear end of L_2 as $(x, y=h)$. The distance between these two points is constant:

$$(x - L_1 \sin \theta)^2 + (h + L_1 \cos \theta)^2 = L_2^2 \quad (2)$$

Therefore,

$$x(\theta) = L_1 \sin \theta - \sqrt{L_2^2 - (h + L_1 \cos \theta)^2} \quad (3)$$

Note this equation implies

$$|h + L_1 \cos \theta| \leq L_2 \quad (4)$$

Otherwise there is no solution.

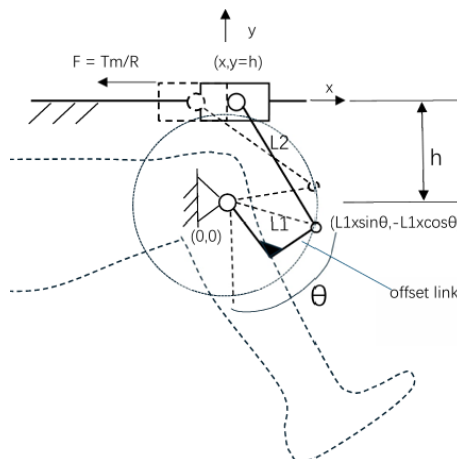


Fig. 5 - Simplified Mechanism Diagram

The derivative of x versus θ is:

$$\frac{dx}{d\theta} = L_1 \cos \theta - \frac{L_1(h + L_1 \cos \theta) \sin \theta}{\sqrt{L_2^2 - (h + L_1 \cos \theta)^2}} \quad (5)$$

Variable L_1 , L_2 , and h values causes different $x(\theta)$ mapping. A preferable configuration is to have $x(\theta)$ monotonic over the operating range, otherwise there would be two solutions for a given θ , which causes mechanical unstability. According to (5), to have $dx/d\theta < 0$,

$$\cos \theta < \frac{(h + L_1 \cos \theta) \sin \theta}{\sqrt{L_2^2 - (h + L_1 \cos \theta)^2}} \quad (6)$$

To ensure monotonic movement ($dx/d\theta < 0$) across the range $\theta \in (0^\circ, 180^\circ)$, the geometric parameters must satisfy:

$$h > (L_2 - L_1) \cos \theta \quad \text{subject to } L_2 \geq |h + L_1 \cos \theta| \quad (7)$$

Note the critical state when $\theta=0$ is lost in calculation. When $\theta=0$ and if $h \neq L_2 - L_1$, $dx/d\theta$ is always positive. $h = L_2 - L_1$ is a critical state that allows $dx/d\theta < 0$ even when $\theta=0$. Realistically the human knee joint is not an ideal hinge with a fixed rotation axis. To have a robust design, it is necessary to have an offset angle to make $\theta_0 > 0$, otherwise for the range between 0° and 90° , some x values may be mapped to two possible θ solutions. One of the implementation method is to have the offset link as shown in Fig. 5 to add in an offset angle of θ . The offset angle safely avoids the θ range that may cause a positive $dx/d\theta$.

In our device, $h = 85\text{mm}$, $L_2 = 200\text{mm}$, the $x(\theta)$ curves for $L_1=110\text{mm}$, 115mm , and 120mm are presented in Fig 6. In the $L_1=115\text{mm}$ case, $h = 85\text{mm} = L_2 - L_1$ is the critical state, in which the $x(\theta)$ shows monotonous $dx/d\theta$ over the interested range of $\theta=0^\circ \sim 150^\circ$. Other two cases both show a positive $dx/d\theta$ at the beginning and the negative when θ is greater than about 18° . For $L_1=120\text{mm}$, the mechanism does not have a solution for $\theta < 16^\circ$.

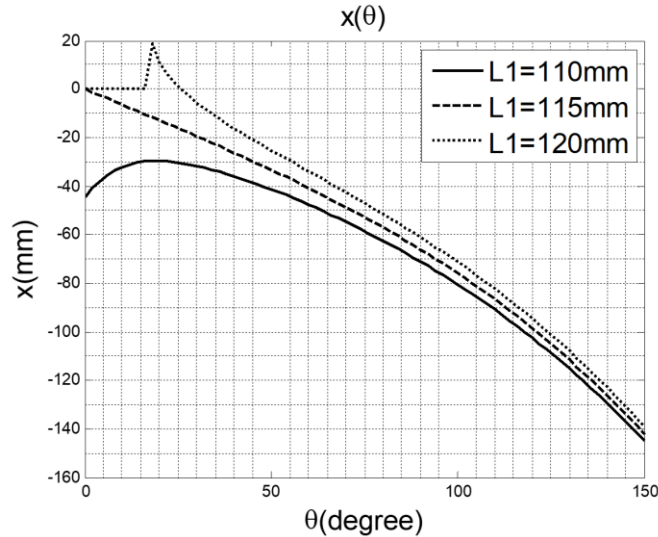


Fig. 6 - Slide-Rotation Relationship

D. Torque analysis

The knee skeleton is driven by a BLDC motor installed on a rack that is attached to the upper thigh. The motor drives a pulley of a radius $R = 17.5\text{mm}$. The wheel then drags through three PU cords on a slide platform. If we assume the torque generated by the BLDC motor is T_m , the force on the slide platform is then $F = T_m/R$. For the L_1 - L_2 link, there is

$$F dx = T d\theta \quad (8)$$

Therefore

$$T = \frac{Fdx}{d\theta} = \frac{T_m dx}{R d\theta} \quad (9)$$

Plug in equation (5) we get the torque on the lower leg

$$T(\theta) = \frac{T_m L_1}{R} \left(\cos \theta - \frac{(h + L_1 \cos \theta) \sin \theta}{\sqrt{L_2^2 - (h + L_1 \cos \theta)^2}} \right) \quad (10)$$

Or equivalently,

$$T_m(\theta) = \frac{R \cdot T}{L_1 \left(\cos \theta - \frac{(h + L_1 \cos \theta) \sin \theta}{\sqrt{L_2^2 - (h + L_1 \cos \theta)^2}} \right)} \quad (11)$$

Equation (11) gives the torque the BLDC should generate when the desired torque on the lower leg is equal to T and the knee joint angle is θ .

IV. Test

A. Test Setup

This knee exoskeleton is 3D printed with nylon. Carbon fiber tubings are used as rails to reduce weight.

Test 1:

The controllable stiffness is tested with a Jitai Universal Testing Machine (UTM), type W5KN. The stiffness is controlled by a laptop, sending angular position commands to the servo motor. The UTM applies a tensile force and measures the resulting displacement of the slide platform relative to the BLDC motor. Then the stiffness is computed with $\Delta F/\Delta L$. The results are shown in Fig. 8.

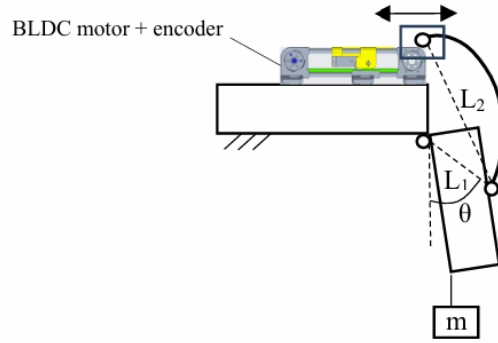


Fig. 7 - In-System Test

Test 2:

In order to further test the effect of the stiffness control inside the system, a 2-piece hinged wood device that mimics human leg is used for the experiment. In the experiment, the BLDC motor is set to drag L_1 and release it from 50 to 0 degrees. The displacement of the slide platform from its $\theta=0$ location is defined as L_s . The axial deformation ΔL of the PU cords is then measured manually by marking along the PU cords. Various stiffness settings are tested to verify the effect. Due to the deformation, Eq. 3. turns into

$$\phi \cdot R = \Delta L + L_1 \sin \theta + \sqrt{L_2^2 - (h + L_1 \cos \theta)^2} \quad (12)$$

ϕ is the rotated angle read from the BLDC motor encoder. R is the radius of the motor pulley. The later part is the lateral position determined by the rigid L_1 - L_2 link. Eq. (12) allows calculation of ΔL based on the motor rotation angle and the slide platform position.

In test 2, the BLDC motor is controlled from a PC to release L1. The motor rotation angle \emptyset is read from an integrated encoder, and the slide platform position is measure manually with a ruler. The elastic deformation of the PU cords is then evaluated to show the performance of the stiffness controller.

B. Test Results

The stiffness is tested by adjusting the position of the slide platform along the PU cords. Due to the unevenness of the PU cords, the stiffness varies over different segments of the cords, and presented a variation of the stiffness for the same cord configuration. Nevertheless, the tested mean stiffness shows a linearity over the whole configuration range.

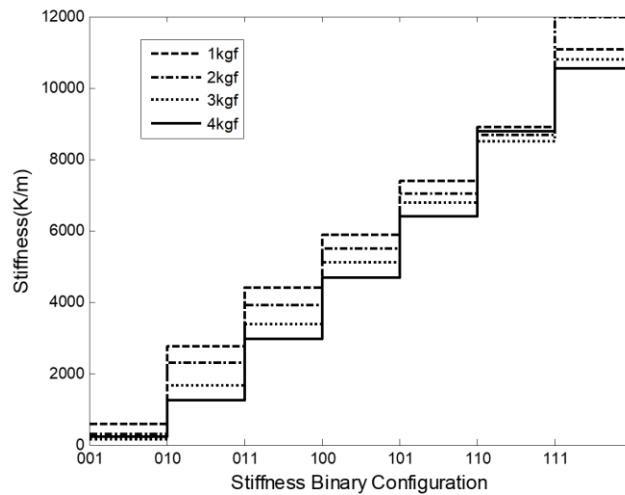


Fig. 8 - Stiffness Measured for Varies PU Cord Configurations

Fig. 8 shows the test results when the cord length is at 10cm, and the load varies from 1kgf to 4kgf. The experiments also shows K decreases with the length between motor pulley and the sliding platform.

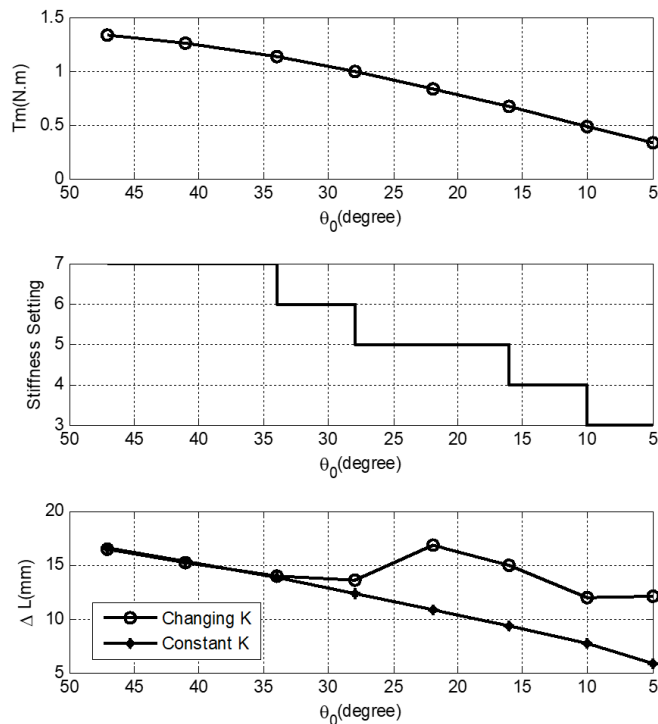


Fig. 9 - In-System Stiffness Test

Fig. 9 shows the data collected as the BLDC motor held L1 and release it from 47 to 0 degrees. The motor has an integrated torque sensor and a angle-sensing encoder. The stiffness is adjusted based on motor torque, with higher stiffness for higher torque. The variation in PU cord cross-section and mechanical backlash may have introduced measurement errors.

The wearable integration of the knee exoskeleton was also evaluated on-body during open-loop joint motion (Fig. 10). The device allowed unhindered articulation across 0 to 90 degrees without mechanical interference.

V. Conclusion

This paper presents a knee exoskeleton actuator featuring digitally controllable variable stiffness based on a binary-coded PU elastic cord architecture. By selectively engaging three elastic cords with binary-weighted cross-sectional areas, the proposed mechanism achieves eight discrete stiffness levels with a linear stiffness relationship. Experimental results demonstrate the feasibility of real-time stiffness modulation and validate the effectiveness of the proposed elastic driver.

The kinematic characteristics of the offset slider-link mechanism were analyzed. By introducing an appropriate offset angle and selecting suitable geometric parameters, the mechanism avoids regions where $dx/d\theta$ becomes positive, improving transmission stability and preventing dual solutions.

The proposed binary-coded stiffness architecture provides a compact and intrinsically compliant approach for exoskeleton applications requiring adaptive torque and stiffness control. Future work will focus on integrating multi-axis inertial sensing and closed-loop control algorithms to enable posture-aware stiffness regulation during human gait.



Fig. 10 - Wearable Evaluation of the Prototype

VI. Acknowledgments

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